An Open Dataset for Activity Recognition in Smart Home Environments Using IMU, Ambient, and Radar Data

Nada Chabih[1], Armel Jacques Nzekon Nzeko'o[1], Florentin Thullier[1], Julien Maître[1], Sébastien Gaboury[1], Kévin Bouchard[1*†]

[1*]Department of mathematics and computer sciences, University of Quebec at Chicoutimi, 555, boulevard de l'Université, Chicoutimi, G7H 2B1, Québec , Canada.

*Corresponding author(s). E-mail(s): kevin_bouchard@uqac.ca;
Contributing authors: nchabih@etu.uqac.ca; ajnnzeko@uqac.ca; fthullie@uqac.ca; jmaitre@uqac.ca; s1gaboury@uqac.ca;
†All authors contributed equally to this work.

Abstract

Technological advancements have catalyzed the emergence of smart homes equipped with connected devices designed to enhance resident comfort and safety. In the context of Ambient Assisted Living (AAL), analyzing telemetry from these devices facilitates robust Human Activity Recognition (HAR), which is essential for the proactive monitoring and support of elderly and disabled populations. However, the field currently faces a shortage of realistic, high-quality datasets capturing diverse Activities of Daily Living (ADL) in smart environments. To address this gap, we developed a smart apartment laboratory outfitted with a multimodal sensor suite, including ambient sensors and radars. We collected synchronized data from these environmental sources alongside Inertial Measurement Unit (IMU) data worn by 19 participants as they performed 14 distinct ADL. Unlike traditional laboratory protocols, data collection occurred in a naturalistic setting where participants engaged in activities freely. This approach captures the inherent variability of human movement, making the dataset highly representative of real-life conditions. We believe this dataset will significantly advance research in HAR for smart habitats, promoting the integration of wearable and contactless sensing technologies in residential settings.

1 Background & Summary

The Internet of Things (IoT) is transforming smart homes by improving security, comfort, and energy efficiency. Devices such as smart thermostats optimize energy consumption [1]. Smart security systems enhance home protection with real-time alerts [2]. In addition, IoT facilitates accessibility for the elderly and disabled, improving autonomous home management [3]. In these smart homes, recognizing human activities is essential, as it improves the interaction between users and IoT systems. To achieve this, analyzing data from sensors, such as cameras, motion sensors, and wearable devices, is important and allows for the detection and interpretation of occupant behaviors in real time [4]. Thus, IoT systems can automatically adapt to the home environment, for example, by adjusting lighting or temperature based on detected activities, thus optimizing comfort and energy efficiency [5]. Moreover, activity recognition can play a crucial role in security, by identifying abnormal behaviors and alerting users or emergency services [6]. Thus, the integration of IoT sensor data analysis for HAR represents a key development to make smart homes more responsive and secure.

Recent research on daily activity recognition in smart homes highlights the importance of collecting diverse data to improve the accuracy and efficiency of the built models. These studies show, for example, that the integration of advanced sensors, such as motion sensors, ambient sensors, cameras, and wearable devices, is crucial to capture a full range of human activities [7, 8]. Moreover, users express a growing need for real-time data and predictive analytics, enabling increased personalization of services offered by smart homes [9, 10]. These advances highlight the need for an integrated approach in data collection to optimize daily activity recognition.

In the work of recognizing daily living activities in smart homes, it is important to have realistic and accurate data to develop effective machine learning models capable of distinguishing human behaviors [11]. This is the case when the use of realistic datasets, including variations in lifestyle habits and social interactions, significantly improves the robustness of activity recognition systems [12]. These results highlight the need to invest in the collection of realistic data to improve the effectiveness and acceptability of activity recognition technologies in smart homes.

Table 1 describes datasets for the recognition of ADL in a smart home, detailing the different types of sensors used and the targeted activities. Their main objective is to recognize simple activities performed by solitary individuals and some by multiple individuals, without disclosing their identities.

Name	Year	Types of sensors used	List of daily living activities	Number of activities	Number of users	Duration (days)
Kasteren et al. (2008)[13]	2008	state-change sensors (14)	Leaving, Toileting, Showering, Sleeping, Breakfast, Dinner, Drink	7	single user	28

Name	Year	Types of sensors used	List of daily living activities	Number of activities	Number of users	Duration (days)
Kasteren et al. (2010)[14]	2010	state-change sensors (14)	Brush teeth, Showering, Toileting, Breakfast, Dinner, Snack, Drink, Load dishwasher, Unload dishwasher	9	single user	25
Kasteren et al. (2010)[14]	2010	state-change sensors (17)	Brush teeth, Showering, Bathing, Shaving, Breakfast, Dinner, Snack, Drink	8	single user	19
Singla et al. (2010)[15]	2010	motion sensors (51), temperature sensors (3), door contacts (12), analog sensors that monitor water and stove burner (3)	Filling medication dispenser, Moving furniture, Watering plants, Playing checkers, Preparing dinner, Reading magazine, Gathering and packing picnic food, Hanging up clothes, Sweeping floor, Setting the table, Paying bills	15	single and two users	
Fleury et al. (2009)[16]	2009	Infra-red presence sensors (6), door contacts (4), temperature sensor (1), hygrometry sensor (1), webcamera (5), microphones (6), and a wearable kinematic sensor (1)	hygiene, toilet use, eating, resting, sleeping, communication, and dressing/undressing	7	single user	
CASA [17]	2012	Infrared motion/light sensor (24), door sensor (1), relay (2), temperature sensor (2)	Bed-toilet transition, Cook, Eat, Enter home, Leave home, Personal hygiene, Phone, Relax, Sleep, Work	10	single user	

Name	Year	Types of sensors used	List of daily living activities	Number of activities	Number of users	Duration (days)
ARAS [18]	2013	force sensitive resistors (3), contact sensors (3), proximity distance sensors (6), photocells (6), temperature sensor (1), and infrared sensors (1)	Sleeping, Watching TV, Studying, Using Internet, Telephone, Preparing breakfast, Having breakfast, Preparing lunch, Having lunch, Preparing dinner, Having dinner, Washing dishes, Having snack, Taking shower, Toileting, Napping, Reading, Brushing teeth, Listening to music, Having conversation, Changing clothes	21 (27 declared)	two users	30
ARAS [18]	2013	force sensitive resistors (3), pressure mat (5), contact sensors (6), proximity distance sensors (4), photocells (2)	Sleeping, Watching TV, Napping, Using Internet, Preparing dinner, Having dinner, Having snack, Taking shower, Toileting, Reading, Shaving, Telephone, Having conversation, Changing clothes	14 (27 declared)	two users	30

Name	Year	Types of sensors used	List of daily living activities	Number of activities	Number of users	Duration (days)
Context-Act@A4H [19]	2017	Water consumption meters (15), Power consumption meters (11), Voltage consumption meters (11), Current consumption meters (11), Other electric-related meters (Energy, frequency, power factor, counter status) (44), Pressure Sensors (1), Contact sensor (21), Curtain Aperture Sensor (8), Lamp State Sensor (17), Smart Electric Outlet (3), Hue Lamp State (5), Switches (54), Thermostat (5), Microphone (7)	Bathing/Showering, Toileting, Feeding/Eating, Mobility, Dressing, Preparing meals, Wash dishes, Laundry, Sleeping, Watching TV, handwashing, brushing teeth, general grooming		single user	60
E-care@home (2017)[20]	2017	light sensor (2), motion sensor (3), temperature sensor, contact sensor, pressure sensor (6), heart beat simulator	in living room, in kitchen, in bathroom, sitting, moving, watching TV, changing pose, exercising, cooking, burning, eating, stressing	12	single user	
McConville et al. (2019)[21]	2019	wearable wrist (1), wireless sensor nodes [gateways] (4)	Livingroom, bedroom, kitchen, Stairs	4	single user, and two users	
E-care@home (2020)[22]	2020	wireless sensor nodes, heart beat simulator, motion sensor (4), pressure sensor (5), light sensor (3), temperature sensor, door sensor	In bathroom, watching tv, sitting, cooking, eating, resting	7	single user, two, three and four users	

Name	Year	Types of sensors used	List of daily living activities	Number of activities	Number of users	Duration (days)
MARBLE [23]	2021	Pressure sensor (9), Cooker smart-plug (1), Television smart-plug (1), Medicine magnetic sensor (1), Pots magnetic sensor (1), Cutlery magnetic sensor (1), Fridge magnetic sensor (1), Pantry magnetic sensor (1)	answering phone, making phone call, clearing table, cooking, eating, entering home, leaving home, preparing cold meal, setting up table, taking medicines, using pc, washing dishes, watching tv	13	single user, two, and four users	
SDHAR-HOME [24]	2022	Motion Sensor (8), Door Sensor (8), Temperature and Humidity Sensor (2), Vibration Sensor (11), Television smart-plug (1), washing machine smart-plug (1), Light Intensity Sensor (2), wearable wrist (2)	Bathroom Activity, Chores, Cook, Dishwashing, Dress, Eat, Laundry, Make Simple Food, Out, Home, Pet, Care, Read, Relax, Shower, Sleep, Take Meds, Watch TV, Work	17	two users	60
Tonkin et al. (2023)[25]	2023	Passive infrared, wearable wrist, cameras, motion sensor, temperature sensor, light sensor, wireless sensor nodes	ascending stairs, descending stairs, jumping, walking, walking with load, bend over, kneeling, lying, sitting on chair/bed/sofa, squatting, standing, bending, straightening, turning/pivoting, stand to kneel, kneel to stand, lying to sit, sit to lying, sit to stand, stand to sit	20	single user	90

Table 1: Datasets for the recognition of activities of daily living ADL in smart homes

To our knowledge, few of these datasets simultaneously consider ambient sensors, ultra-wideband radars, motion sensors, cameras, and smartwatches in a real-time, participant-centric data collection process without a predefined sequence of activities to be performed.

We present a dataset collected from 19 participants who performed 14 different human activities in a prototype smart apartment designed by the Ambient Intelligence for Activity Recognition Laboratory (LIARA). Data collection was conducted using IMU wristbands measuring movement signals along the X, Y, and Z axes at a frequency

of 50 Hz. The raw data files are stored in JSON format and contain participant IDs, activity names, and the rooms in which they took place.

The approach used to obtain this dataset is authentic and participant-centered. The data is collected in a realistic environment, allowing participants to perform activities freely, rather than following predefined and restrictive sequences. This enhances the natural variability of movement patterns, making the dataset more reminiscent of real-life conditions and improving its applicability to practical smart home environments.

2 Methods

2.1 Dataset design.

In this section, we provide a guide for acquiring data from different sensors. The collected data represents ADL that a person can perform in a smart home (a house or apartment). To do this, LIARA has an apartment augmented with different sensors that provide data during the simulation of daily living activities by previously recruited participants. We also express rules to be respected in order to ensure good integrity in the standardization of the data that is collected. In addition, these rules will also simplify data manipulation (data pre-processing) during the ADL recognition process (for example by machine learning algorithms), and ensure minimal, if not zero, data loss.

The datasets presented in this paper are collected in two different contexts: the single-resident context and the multi-resident context which will be described separately in the rest of the document.

2.2 Data collection setup.

2.2.1 Data collection in single-resident context.

Regarding the data collection process in a single-resident context, the sensors used are given by the following list:

- All ambient sensors historically present in the laboratory:
  - Pressure sensors
  - Temperature sensors
  - Light sensors
  - Water flow sensors
  - Water temperature sensors
  - Motion sensors
  - Electromagnetic switches
- 03 ultra-wideband (UWB) radars
- 01 Bluetooth wristband with a 6-axis inertial measurement unit (IMU) (3 accelerometric axes and 3 gyroscopic axes)
- 02 depth cameras (RGBD)

All sensors are static, meaning that, with the exception of the Bluetooth wristbands, they are arranged in the LIARA laboratory according to the plan illustrated in

Figure 1. Note that additional ambient sensors are present in the apartment, though they are omitted from this diagram for clarity.

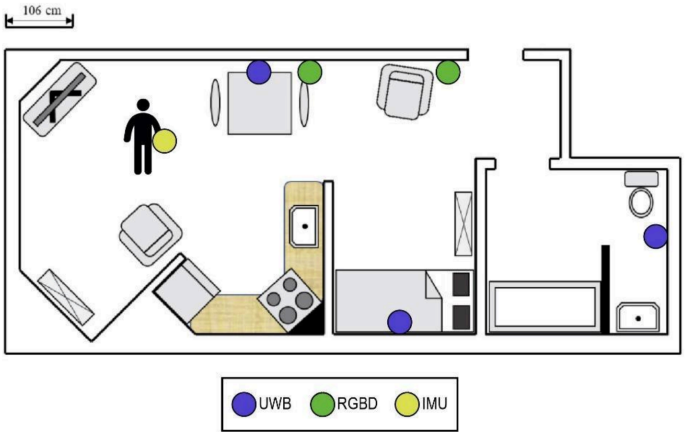


Fig. 1 Laboratory plan showing the location of the main sensors for single-resident data collection, where the blue circles represent the ultra-wideband radars, the green circles identify the depth cameras, and the yellow circle indicates the Bluetooth wristband.

As shown in Figure 2, the wristband's physical address should be positioned as if it were indicating 6 o'clock on a wristwatch.

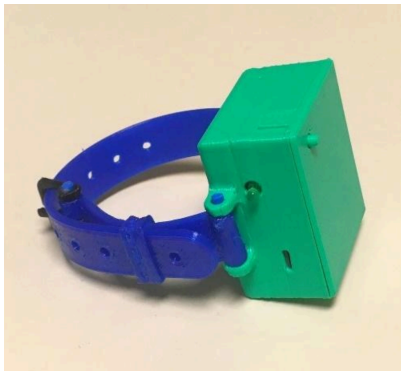


Fig. 2 Bluetooth wristband used for data collection.

Data is collected on various servers that distribute data from ambient sensors, ultra-wideband sensors, and the Bluetooth wristband via the WebSocket protocol. All this is possible thanks to the smart apartment's private Wi-Fi network.

2.2.2 Data collection in multi-resident context.

Regarding the data collection procedure in a multi-resident setting, in addition to the sensors used in the single-resident setting, we also use:

- 03 PIR+ motion sensors, which also allow indoor localization via Bluetooth signal strength (RSSI) when used in combination with the wristbands
- A second Bluetooth wristband for the second participant

With the exception of the Bluetooth wristbands, all sensors are static and are arranged in the LIARA laboratory according to the layout shown in Figure 3. The PIR+ sensors are positioned on the ceiling. The Bluetooth wristband must be placed on the wrist of each participant's dominant hand. As with the single-resident data collection, the physical address of the wristband must be positioned as if it indicated 6 o'clock on a watch, and the wristband bearing the letter A must be assigned to participant 1 (P1) and the one bearing the letter B to participant 2 (P2).

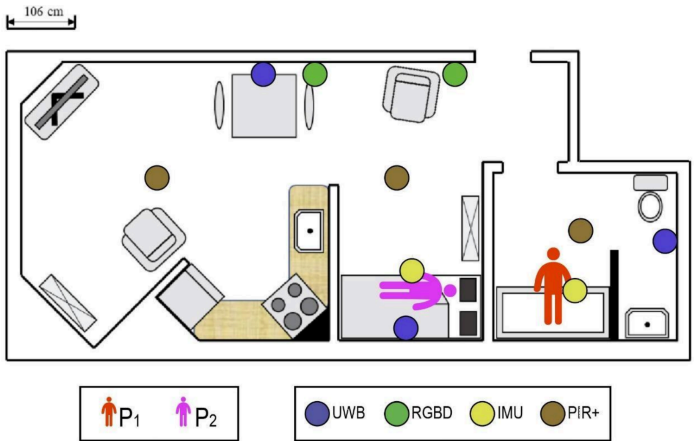


Fig. 3 Laboratory plan showing the location of the main sensors for multi-resident data collection, where blue circles show ultra-wideband radars, green circles identify depth cameras, yellow circles indicate Bluetooth wristbands, and brown circles represent PIR+ sensors. In addition, the two participants P1 and P2 are identified in red and purple, respectively.

2.3 Sensor Synchronization and Data Streaming Protocol

Data from the sensor network is streamed in real time using the WebSocket protocol over the LIARA smart apartment's private Wi-Fi network (SSID: Liara) [2, 9]. This setup handles data streams from all sources concurrently: the three static ultra-wideband (UWB) radars, two depth cameras (RGBD), the 6-axis Bluetooth Inertial Measurement Unit (IMU) wristbands sampling at 50 Hz, and the built-in ambient sensor grid (tracking pressure, temperature, light, motion, water flow, water temperature, and electromagnetic switch states) [8, 24, 25]. Each sensor category transmits data directly to its own server instance.

Because these devices use distinct sampling rates and data modalities, incoming data packets are captured as structured JSON files to help align the signals for multi-sensor fusion [4, 7]. The generated files use a strict naming convention that logs the participant's unique anonymous ID, the sensor type, the activity label, the iteration number, and the room location [22]. To ensure all activities are fully captured and to

avoid signal loss during start-up, every recording begins exactly 10 seconds before the participant starts the simulated task and ends 5 seconds after the task is finished. This padding provides a consistent temporal baseline across all parallel channels, simplifying signal alignment and event segmentation during data preprocessing [12].

2.4 Data annotation procedure.

2.4.1 Activity identification.

For each activity performance, the recording begins 10 seconds before the participant begins simulating the activity. In addition, the recording continues for 5 seconds after the participant completes the activity.

In the case of multi-resident data, two participants simultaneously perform two activities in the apartment. We consider a total of 9 distinct activity combinations grouped into three categories. Just as in the single-resident case, the sequence of activities closely resembles everyday situations. The list of all activity combinations is described in Table 2.

		Same piece Same activity	**Same piece Different activities**	**Different pieces Different activities**
Combination 1	Participant 1 Participant 2	Fold the laundry Fold the laundry		
Combination 2	Participant 1 Participant 2	Eat Eat		
Combination 3	Participant 1 Participant 2	Brush the teeth Brush the teeth		
Combination 4	Participant 1 Participant 2		Wash the dishes Eat	
Combination 5	Participant 1 Participant 2		Put away the dishes Watch TV	
Combination 6	Participant 1 Participant 2		Brush the teeth Take a shower	
Combination 8	Participant 1 Participant 2			Get dressed Go to the bathroom
Combination 9	Participant 1 Participant 2			Sleep Watch TV

Table 2 List of activity combinations performed during the multi-resident data collection

Single-resident data collection involves recording a total of 14 distinct activities. These activities are performed freely by each participant, one after the other, in the LIARA apartment, and their sequence closely resembles everyday situations. The list of all activities is as follows:

- Brushing teeth
- Taking a shower
- Going to the bathroom
- Sleeping
- Dressing/undressing

- Housework
- Washing dishes
- Putting away dishes
- Resting
- Using a computer (or cell phone)
- Putting away laundry
- Eating
- Drinking
- Reading

2.4.2 Activity Recording in the single-resident case.

In this subsection, we describe how each activity was recorded in the single-resident context. Table 3 shows the duration of each activity, the number of times it was recorded, and finally the location where the activity was performed.

1. **Brushing teeth.** This activity takes place twice in the bathroom, and the participant may not leave the room. Before the recording began, the participant stood in front of the sink. In addition, the toothbrush, provided by the laboratory, must already be placed in the cup provided for this purpose, which is located on the shelf above the sink. During the recording, the participant can use the sink, brush their teeth, rinse their mouth, and use their toothbrush. The researcher responsible for the data collection notified the participant 15 to 20 seconds before the end of the recording so that they can take the necessary steps to complete the activity.

2. **Taking a shower.** This activity takes place twice in the bathroom, and the participant may not leave the room. Before the recording begins, the participant must remove their shoes and stand on the mat in front of the bathtub. During the recording, the participant may, for example, simulate opening and closing the toilet, washing their hair and body, rinsing, etc. The researcher responsible for

Activities	Duration	Number of Iterations	Location
Brush the teeth	1 minute +15s	2	Bathroom
Take a shower	2.5 minutes +15s	1	Bathroom
Go to the bathroom	1 minute + 15s	2	Bathroom
Sleep	2 minutes + 15s	1	Bedroom
Getting dressed/undressed	45 seconds + 15s	2	Bedroom
Do the cleaning	1 minute + 15s	2	Choose a place
Wash the dishes	2 minutes + 15s	1	Kitchen
Put away the dishes	1 minute + 15s	2	Kitchen
Rest	1 minute + 15s	2	Choose a place
Use a computer / cellphone	2 minutes + 15s	1	Choose a place
Put away the laundry	2 minutes + 15s	1	Bedroom
Eat	2 minutes + 15s	1	Dining room
Drink	2 minutes + 15s	2	Choose a place
Read	2 minutes + 15s	1	Choose a place

Table 3 Description of the progress of recordings of each of the 14 activities considered in the single-resident case.

the data collection notified the participant 20 to 30 seconds before the end of the recording so that they can perform the necessary actions to complete the activity.

3. **Going to the bathroom.** This activity takes place twice in the bathroom, and the participant may not leave the room. Before the recording begins, the participant must stand facing or facing away from the toilet, depending on how it will be used (sitting or standing). During the recording, the participant may, for example, simulate handling their pants, flushing the toilet, wiping, etc. The researcher in charge of the data collection must notify the participant 15 to 20 seconds before the end of the recording so that the participant can perform the necessary actions to complete the activity.

4. **Sleeping.** This activity takes place only once in the bedroom, and the participant may not leave the room. Before the recording begins, the participant must stand facing the bed. During the recording, the participant can simulate falling asleep by moving in bed, or simulate being deeply asleep, meaning they will not move. The researcher in charge of the data collection must take notes on the participant's activity throughout the recording.

5. **Dressing/undressing.** This activity takes place twice in the bedroom, and the participant may not leave the room. Before the recording begins, the participant must stand with their back to the bed. During the recording, the participant can, for example, take off their shoes (or put them back on), get dressed to go outside (or undress after coming back from outside).

6. **Housework.** This activity takes place twice, and the participant cannot leave the selected room for each iteration. During the recording, the participant can only perform one proposed task at a time. For the first iteration, the possible tasks are:

   - Clean the kitchen counter
   - Clean the kitchen table

   To do this, the participant can use the bottle of cleaning product and the paper provided. For the second iteration, the proposed tasks are as follows:

   - Vacuum the living room
   - Vacuum the bedroom
   - Vacuum the kitchen
   - Sweep the living room and pick up the dust with the dustpan
   - Sweep the bedroom and pick up the dust with the dustpan
   - Sweep the kitchen and pick up the dust with the dustpan

   To do this, the participant may use the broom, dustpan, or vacuum cleaner provided next to the living room armchair. Furthermore, the researcher in charge of the data collection must take notes on the participant's activity, for each iteration, throughout.

7. **Washing dishes.** This activity takes place only once in the kitchen, and the participant may not leave this room. Before the recording began, the participant stood in front of the sink. During the recording, the participant must wash dishes that have been previously taken out by the researcher in charge of the data collection. To do this, the participant may use a sponge and dish soap provided on the kitchen counter.

8. **Putting away dishes.** This activity takes place twice in the kitchen, and the participant cannot leave this room. Before the recording began, the participant stood in front of the sink. During the recording, the participant must put away the previously washed dishes. To do this, the participant can use the cloth provided on the oven handle. The researcher responsible for the data collection must indicate to the participant which cupboards the clean dishes should be stored in.

9. **Resting.** This activity takes place twice, either in the living room or the bedroom, and the participant cannot leave the room chosen for each iteration. The participant begins the recording already seated in the armchair provided in the room in question. Given the sensors available for this data collection, we will not be able to properly differentiate between resting and watching TV. Therefore, the participant may watch TV while resting in the armchair in the living room.

10. **Using a computer (or cell phone).** This activity takes place only once, and the participant cannot leave the room chosen for each iteration. The participant begins the recording already in position and can choose to use their cell phone or a (non-functional) computer provided. Regarding locations, the participant is offered the following options:

    - Use their cell phone or computer while sitting at the kitchen table
    - Use their cell phone or computer while sitting in the living room armchair
    - Use their cell phone or computer while sitting in the bedroom armchair
    - Use their cell phone while standing, leaning on the kitchen counter

    The researcher responsible for the data collection must take notes on the participant's activity throughout the recording (sitting or standing position, accessories used, etc.).

11. **Putting away laundry.** This activity takes place only once in the bedroom, and the participant may not leave the room. Before the recording begins, the participant must stand facing the bed. During the recording, the participant must fold the laundry provided into a basket and then store it in the wardrobe next to the bed.

12. **Eating.** This activity takes place only once in the dining room, and the participant may not leave the room. The participant begins the recording already in position, eating from the serving set that will have been placed on the table in advance by the researcher in charge of the data collection.

13. **Drinking.** This activity takes place twice, and the participant may not leave the room chosen for each iteration. The participant begins the recording already in position and may choose to drink a cup of coffee or tea, a glass of water, a bottle of water, etc. The container must, however, be filled beforehand, whether the participant brings their own liquid or has it provided by the researcher in charge. Regarding the locations, the participant is offered the following choices:

    - Drinking while seated in the living room armchair
    - Drinking while seated in the bedroom armchair
    - Drinking while standing, leaning on the kitchen counter

    The researcher responsible for the data collection must take notes on the participant's activity throughout the recording (sitting or standing position, speed of execution, etc.).

14. **Reading.** This activity is performed only once, and the participant may not leave the selected room. The participant must. The participant begins the recording already in position and can choose to read a book provided to them while sitting in the living room armchair or the bedroom armchair.

2.4.3 Activity Recording in the multi-resident case.

In this subsection, we describe how each activity combination was recorded in the multi-resident context. Table 4 shows the duration of each combination, the number of times it was recorded, and finally the location where the activities were performed by the first and second participants.

Combinations	Duration	Number of Iterations	Location P1	Location P2
Combination 1	2 minutes +15s	1	Bedroom	Bedroom
Combination 2	2 minutes +15s	1	Dining room	Dining room
Combination 3	1.5 minute + 15s	2	Bathroom	Bathroom
Combination 4	2 minutes + 15s	1	Kitchen	Dining room
Combination 5	1 minute + 15s	2	Kitchen	Living room
Combination 6	1.5 minute + 15s	2	Bathroom	Bathroom
Combination 7	2 minutes + 15s	1	Bathroom	Bedroom
Combination 8	1 minute + 15s	2	Bedroom	Bathroom
Combination 9	1 minute + 15s	2	Bedroom	Living room

Table 4 Description of the progress of recordings of each of the nine combinations considered in the multi-resident case.

1. **Combination 1.** This activity takes place only once in the bedroom, and participants may not leave the room. Before the recording begins, both participants must stand facing the bed. During the recording, participants must fold the laundry provided in a basket and then put it away in the wardrobe next to the bed. Participants can organize themselves as they see fit. For example, one can fold the laundry while the other puts it away, or both can fold and put it away at the same

time, etc.

2. **Combination 2.** This activity takes place only once in the dining room, and participants may not leave the room. Participants begin the recording already in position, eating from the utensils that will have been placed on the table beforehand by the researcher in charge of the data collection.

3. **Combination 3.** This activity takes place twice in the bathroom, and participants may not leave the room. Before the recording begins, participants must stand more or less close to the sink. Additionally, the toothbrushes provided by the laboratory must be placed in the designated cup located on the shelf above the sink. During the recording, participants can use the sink and brush their teeth simultaneously, rinse their mouths in turns, etc. The researcher in charge of the data collection must notify participants 20 to 30 seconds before the end of the recording so that they can take the necessary steps to complete the activity.

4. **Combination 4.** This activity is conducted only once, and participants may not leave their respective rooms. Participant 1 (P1) is in the kitchen, and Participant 2 (P2) is in the dining room. Before the recording begins, P1 must stand in front of the sink, while P2 begins the recording already in position, eating from the serving set that will have been placed on the table in advance by the researcher in charge of the data collection. During the recording, P1 must wash dishes that will have been taken out beforehand by the researcher. To do this, the participant can use a sponge and dish soap provided on the kitchen counter.

5. **Combination 5.** This activity takes place twice, and participants are not permitted to leave their respective rooms. Participant 1 (P1) is in the kitchen, and Participant 2 (P2) is in the living room. Before the recording begins, P1 must stand in front of the sink, while P2 begins the recording seated in the living room armchair. During the recording, P1 must put away the previously washed dishes. To do this, the participant can use the cloth provided on the oven handle. The researcher in charge of the data collection must indicate to the participant which cupboards the clean dishes should be stored in.

6. **Combination 6.** This activity takes place twice, and participants are not permitted to leave their respective rooms. Both participants are in the bathroom. Before the recording begins, P1 must stand in front of the sink, while P2 must have removed their shoes and stand on the mat in front of the bathtub. The toothbrush used by P1, which is provided by the laboratory, must already be placed in the designated cup located on the shelf above the sink. During the recording, P1 can use the sink, brush their teeth, rinse their mouth, and use their toothbrush. P2, for example, can simulate opening and closing the sink, washing their hair and body, rinsing, etc. The researcher responsible for the data collection must notify both participants 20 to 30 seconds before the end of the recording so that they can perform the necessary actions to complete the activity.

7. **Combination 7.** This activity is conducted only once, and participants are not permitted to leave their respective rooms. Participant P1 is in the bathroom, and participant P2 is in the bedroom. Before the recording begins, P1 and P2 must have removed their shoes, and P1 must stand on the mat in front of the bathtub, while P2 must stand in front of the bed. During the recording, P1 can, for example, simulate opening and closing the shower, washing their hair and body, rinsing, etc. P2 can simulate falling asleep by moving in bed, or being deeply asleep, meaning they will not move at all. The researcher in charge of the data collection must notify the participants 20 to 30 seconds before the end of the recording so that P1 can make the necessary movements to complete the activity. Furthermore, it is important to take notes on P2's activity throughout the recording.

8. **Combination 8.** This activity takes place twice, and participants are not permitted to leave their respective rooms. Participant P1 is in the bedroom, and participant P2 is in the bathroom. Before the recording begins, P1 must stand with his back to the bed, while P2 must stand facing or facing away from the toilet, depending on how it will be used (sitting or standing). During the recording, P1 can, for example, take off his shoes (or put them back on), get dressed to go outside (or undress after coming back from outside). P2 can simulate handling pants, flushing the toilet, wiping, etc. The researcher responsible for the data collection must notify participants 15 to 20 seconds before the end of the recording so that P2 can perform the necessary actions to complete the activity.

9. **Combination 9.** This activity takes place twice, and participants are not permitted to leave their respective rooms. Participant (P1) is in the bedroom and participant 2 (P2) is in the living room. Before the recording begins, P1 must have removed his shoes and stand facing the bed, while P2 begins the recording already sitting in the living room armchair. During the recording, P1 simulates falling asleep by moving in bed or being deeply asleep, that is, he will not make any movement. The researcher responsible for the collection must take notes on P1's activity throughout the recording.

3 Data Records

3.1 Naming for data files.

For data collection in a single-resident context, each recording must produce four separate data files, one per sensor type. Therefore, the names of the resulting files must comply with the following nomenclature:
[participant-ID]_[sensor]_[activity-label]_[iteration-number]_[location].[extension]

For example, for participant David Gagnon, whose unique ID is 8157ccdf, who simulated the drinking activity for the second time while leaning on the kitchen counter, the data from the UWB radars will be stored in the file named:

8157ccdf_UWB_Drinking_2_Kitchen.json

For each participant, a unique ID must be generated to ensure the confidentiality of their data. Furthermore, for collection in the multi-resident context, there will be 2 additional files: one for the PIR+ and a second file for the data from the Bluetooth bracelet of participant 2, making 6 in total. Therefore, for this collection, the names of the resulting files must comply with the following nomenclature:
[participant-ID]_[sensor]_[P1 or P2]_[Combination]_[iteration-number]_[location].[extension]

For example, for participant David Gagnon, whose unique identifier is 8157ccdf and who is participant 2, having simulated activity combination number 8, as described above, for the first time, the data from his Bluetooth bracelet will be stored in the file whose name must be:
8157ccdf_IMU_P2_Combination8_1_Bathroom.json

The list of possibilities for each of the blocks that make up the naming convention is given below:

- Sensors: AmbientSensors, IMU, UWB, RGBD, PIR
- Activity Labels: BrushingTeeth, DoingHousework, Drinking, Eating, GettingDressedUndressed, GoingToilet, PuttingAwayCleanLaundry, PuttingAwayDishes, ReadingBook, Resting, Sleeping, TakingShower, UsingCompSPhone, WashingDishes
- Locations: Kitchen, Livingroom, Bedroom, Diningroom, Bathroom

3.2 Organization of data files.

To ensure accessibility and facilitate efficient data loading for machine learning pipelines, the dataset follows a hierarchical organization based on the context of the activity. This structure is designed to be intuitive, allowing for programmatic access through standard file-system traversal, which minimizes the overhead required for researchers to partition data into training, validation, and test sets.

In the single-resident context, it is necessary to respect the tree structure defined as follows:

mono (folder)
|
participant_id (folder)
|
activity_label (folder)
|
data files for each iteration

For data collection in a multi-resident context, it is necessary to respect the tree structure defined as follows:

multi (folder)
|
combination (folder)
|
data files for each participant and for each iteration

This standardized tree structure not only ensures data consistency across the multi-resident and single-resident scenarios but also simplifies the development of automated data loaders. By embedding metadata, such as participant ID, activity, and location, directly into the file naming convention and directory hierarchy, the repository remains highly scalable and readily compatible with diverse HAR training frameworks.

4 Technical Validation

To demonstrate the synchronization fidelity and predictive utility of the compiled multimodal repository, we conducted a baseline technical validation evaluating both the contactless ultra-wideband (UWB) radar and the kinematic wearable (IMU) modalities.

4.1 Contactless Radar Validation (UWB)

We evaluated deep convolutional neural network (CNN) architectures under three distinct spatial constraints: whole-apartment classification (C_1), proximity-based signal routing (C_2), and per-room isolated classification (C_3) [4] as shown in table 5. All model variations were thoroughly verified using a rigorous Leave-One-Subject-Out (LOSO) cross-validation framework across 190 experimental cycles, ensuring complete decoupling of participant identities between the training, validation, and testing partitions to eliminate identity leakage [11].

Spatial Strategy	Environment Context	Optimal Window	Peak Baseline F1-score
Apartment-Level (C_1)	Global Footprint	10 sec	0.798 ± 0.023
Position-Based (C_2)	Dynamic Proximity	15 sec	0.798 ± 0.031
Room-Level (C_3)	Bathroom	5 sec	1.000 ± 0.000
Room-Level (C_3)	Dining Room	15 sec	0.997 ± 0.002
Room-Level (C_3)	Kitchen	5 sec	0.875 ± 0.020
Room-Level (C_3)	Living Room	15 sec	0.572 ± 0.092
Room-Level (C_3)	Bedroom	15 sec	0.572 ± 0.040

Table 5 Baseline Technical Validation Benchmarks Using Optimal Preprocessing Methods

The empirical benchmarks demonstrate that signal smoothing via moving average preprocessing is vital to suppress ambient clutter and static background reflections across extended tracking periods [12]. In the global apartment layout (C_1), the moving average pipeline maximized predictive stability at a 10-second temporal window, achieving a Top-1 classification accuracy of 0.799 ± 0.025 and a balanced accuracy of 0.807 ± 0.025.

When transitioning to localized topologies, the room-level strategy (C_3) consistently outperformed global strategies by leveraging room boundaries as spatial priors, narrowing candidate behavioral classes and sharpening classification decision boundaries. Highly distinct kinematic routines within isolated environments produced near-perfect recognition metrics; the bathroom-specific model reached a baseline accuracy and F1-score of 1.000 ± 0.000 using a 5-second window, while the dining room model peaked at 0.998 ± 0.001 using a 15-second moving average filter. Conversely, environments featuring subtle micro-movements or highly transitional resting postures presented greater classification complexity, establishing baseline accuracies of 0.742 ± 0.044 in the bedroom and 0.742 ± 0.056 in the living room under 15-second windows. These baseline achievements validate the underlying synchronization of the data collection pipeline and confirm that the dataset is highly suitable for building advanced multimodal sensor-fusion models.

4.2 Kinematic Wearable Validation (IMU)

To verify the quality and practical utility of the kinematic wearable data, a baseline technical validation was performed targeting the 6-axis IMU wristband telemetry.

The kinematic wristband data stream (sampling at 50 Hz) was validated using four interpretable, computationally efficient machine learning frameworks: K-Nearest Neighbors (KNN), Random Forest, XGBoost, and CatBoost [8, 24, 25]. Preprocessing matched the temporal windowing strategies used for the radar streams, with high-frequency noise suppressed via Chebyshev filtering. Feature extraction yielded 42 statistical and frequency attributes per time-series segment, optimized through feature selection with SelectKBest and Principal Component Analysis (PCA) [22].

Validation was conducted across two distinct spatial constraints: global apartment-level classification (C_1) and per-room isolated classification (C_3). All model configurations were systematically verified using a rigorous Leave-One-Subject-Out (LOSO) cross-validation framework to ensure complete decoupling of participant identities between training, validation, and testing partitions.

Spatial Strategy	Environment Context	Optimal Window	Peak Baseline F1-score
Apartment-Level (C_1)	Global Footprint	15 sec	0.602
Room-Level (C_3)	Kitchen	15 sec	0.930
Room-Level (C_3)	Bathroom	15 sec	0.880
Room-Level (C_3)	Dining Room	15 sec	0.800
Room-Level (C_3)	Bedroom	15 sec	0.680
Room-Level (C_3)	Living Room	5 sec	0.500

Table 6 Baseline Technical Validation Benchmarks Using Optimal Preprocessing Methods for IMU Wristbands

In flat, apartment-wide classification configurations (C_1) lacking localized spatial contexts, ensemble tree models achieved an overall baseline F1-score of 60.21% (Random Forest) and 60.00% (XGBoost) over a 15-second window. While highly dynamic routines like *Brushing Teeth* maintained high precision, matching F1 benchmarks of

89.00% across architectures, mapping individual behaviors globally over the whole apartment introduced significant inter-class ambiguity; these comparative baselines are summarized in Table 6. Structurally similar tasks were frequently misclassified, such as *Eating* as *Drinking*, or *Sleeping* as *Resting* [14, 15, 18].

To break these ambiguity limits, two validation strategies were implemented. First, integrating the room location directly as a categorical attribute within the global feature matrix yielded an exceptional performance leap, elevating overall classification accuracy to a range of 73.00% - 78.00% while lifting the lowest individual activity F1-score from 20.00% to 49.00%.

Second, transitioning to localized topologies (C_3) dramatically lowered error rates by eliminating cross-activity confusions and acting as a spatial prior. Highly distinct kinematic routines achieved robust classification baselines over 15-second windows; notably, a localized bathroom sub-model reached an average F1-score of 88.00% for Random Forest and CatBoost models, explicitly optimizing the segmentation of *Brushing Teeth* (93.00%), *Going Toilet* (84.00%), and *Taking Shower* (87.00%). Similarly, localized kitchen configurations using PCA and Chebyshev filtering generated stable peak F1-scores of 0.930, completely resolving the global model confusions between *Washing Dishes* and *Putting Away Dishes*. Conversely, environments characterized by subtle micro-movements, high class-imbalance, or minimal movement profiles established lower peak baselines, such as 0.680 in the bedroom and 0.500 in the living room.

These multimodal baseline configurations validate the integrity of the IMU data repository and prove its mathematical viability for training state-of-the-art sensor fusion and context-aware activity recognition systems [7].

Ethics and Privacy Statements

The data collection protocol involving human subjects was formally reviewed and approved by the institutional ethics review board of the Université du Québec à Chicoutimi (UQAC). Prior to commencing the experimental sessions, informed consent was obtained from all 19 participants recruited for the study. Participants were fully briefed on the nature of the sensor suite, the duration of the recording sessions, and their right to withdraw from the study at any point without penalty. Because the data acquisition took place in a naturalistic smart apartment setting containing highly sensitive domestic zones, specifically the bedroom and bathroom, where participants freely simulated intimate ADL [13], a strict privacy-by-design pipeline was enforced throughout the data lifecycle. To guarantee complete participant confidentiality, all recorded data streams are entirely decoupled from real-world identities and mapped strictly to randomly generated, unique alphanumeric participant identifiers (e.g., `8157ccdf`).

While this multimodal dataset integrates advanced environmental tracking, comprising ultra-wideband (UWB) radars, ambient sensors, and wearable inertial measurement units (IMUs), it also incorporates two depth cameras (RGBD). To mitigate the unique privacy risks introduced by optical sensing in private spaces, these RGBD cameras were configured to capture structural depth maps rather than high-resolution

color video imagery, preventing the recording of explicit facial features, skin tones, or identifiable visual profiles [4]. Furthermore, in multi-resident contexts where two individuals simultaneously interact within the same rooms, data isolation is maintained by partitioning synchronous signals into independent, anonymized files per participant to eliminate the risk of unauthorized cross-identity behavioral profiling or signal confusion in multi-inhabitant tracking layouts [15, 23].

Beyond initial anonymization, we have implemented rigorous data governance protocols to manage the repository's long-term utility and security. Access to the raw telemetry is restricted to authorized research personnel, and the data is stored on secure, encrypted infrastructure managed by the LIARA laboratory. This governance framework ensures that the dataset remains a resource for academic advancement while protecting the participants from potential re-identification or misuse. By providing a clean, curated, and standardized dataset, we aim to minimize the need for external researchers to access the raw, unanonymized source files, thereby reducing the secondary exposure of sensitive behavioral telemetry.

The primary societal and clinical objective of this research repository is to empower the development of robust HAR models for AAL and remote health monitoring of vulnerable populations [3, 10]. However, the authors acknowledge that high-precision indoor localization and continuous multi-sensor tracking carry inherent data surveillance risks if repurposed for unauthorized monitoring or domestic intrusion [2]. Consequently, future real-world deployments utilizing architectures optimized on this dataset must strictly mandate secure encryption protocols and edge-processing paradigms to prevent raw signal transmission outside the home, establishing a foundational footprint for secure smart-home activity recognition architectures [6].

Looking forward, we emphasize the necessity of addressing "algorithmic fairness" in the context of smart home monitoring. As HAR models become more deeply integrated into clinical and residential health systems, it is vital to ensure that these systems perform equitably across diverse populations. Finally, continuous expansion of demographic diversity, including participants of varying age groups, body masses, and physiological mobility levels, remains a core priority to systematically mitigate algorithmic bias and ensure that AAL technologies provide reliable support to all individuals regardless of their physical or demographic characteristics. This dataset is intended to serve as a catalyst for such research, promoting a future where smart home technologies are as inclusive as they are innovative.

References

[1] Wolniak, R., Grebski, W.: The usage of smart thermostats in smart home. Zeszyty Naukowe Politechniki Śląskiej. Organizacja i Zarzadzanie **189**, 759–769 (2023)

[2] Pandya, S., Ghayvat, H., Kotecha, K., Awais, M., Akbarzadeh, S., Gope, P., Mukhopadhyay, S.C., Chen, W.: Smart home anti-theft system: a novel approach for near real-time monitoring and smart home security for wellness protocol. Applied System Innovation **1**(4), 42 (2018)

[3] Pal, D., Triyason, T., Funikul, S.: Smart homes and quality of life for the elderly: a

systematic review. In: 2017 IEEE International Symposium on Multimedia (ISM), pp. 413–419 (2017). IEEE

[4] Bouchabou, D., Nguyen, S.M., Lohr, C., LeDuc, B., Kanellos, I.: A survey of human activity recognition in smart homes based on iot sensors algorithms: Taxonomies, challenges, and opportunities with deep learning. Sensors **21**(18), 6037 (2021)

[5] Anvari-Moghaddam, A., Monsef, H., Rahimi-Kian, A.: Optimal smart home energy management considering energy saving and a comfortable lifestyle. IEEE Transactions on Smart Grid **6**(1), 324–332 (2014)

[6] Palagan, C.A., Raj, T.S.R., Pillai, N.M., Yamini, K.A.P.: Ssars: Secure smart-home activity recognition system. Computers and Electrical Engineering **123**, 110203 (2025)

[7] Ding, J., Wang, Y., Si, H., Gao, S., Xing, J.: Multimodal fusion-adaboost based activity recognition for smart home on wifi platform. IEEE sensors journal **22**(5), 4661–4674 (2022)

[8] Bianchi, V., Bassoli, M., Lombardo, G., Fornacciari, P., Mordonini, M., De Munari, I.: Iot wearable sensor and deep learning: An integrated approach for personalized human activity recognition in a smart home environment. IEEE Internet of Things Journal **6**(5), 8553–8562 (2019)

[9] Perumal, T., Ramanujam, E., Suman, S., Sharma, A., Singhal, H.: Internet of things centric-based multiactivity recognition in smart home environment. IEEE Internet of Things Journal **10**(2), 1724–1732 (2022)

[10] Talal, M., Zaidan, A., Zaidan, B., Albahri, A.S., Alamoodi, A.H., Albahri, O.S., Alsalem, M., Lim, C.K., Tan, K.L., Shir, W., *et al.*: Smart home-based iot for real-time and secure remote health monitoring of triage and priority system using body sensors: Multi-driven systematic review. Journal of medical systems **43**(3), 42 (2019)

[11] Kulsoom, F., Narejo, S., Mehmood, Z., Chaudhry, H.N., Butt, A., Bashir, A.K.: A review of machine learning-based human activity recognition for diverse applications. Neural Computing and Applications **34**(21), 18289–18324 (2022)

[12] Wan, J., O'grady, M.J., O'Hare, G.M.: Dynamic sensor event segmentation for real-time activity recognition in a smart home context. Personal and Ubiquitous Computing **19**(2), 287–301 (2015)

[13] Van Kasteren, T., Noulas, A., Englebienne, G., Kröse, B.: Accurate activity recognition in a home setting. In: Proceedings of the 10th International Conference on Ubiquitous Computing, pp. 1–9 (2008)

[14] Van Kasteren, T., Englebienne, G., Kröse, B.J.: Activity recognition using semi-markov models on real world smart home datasets. Journal of ambient intelligence and smart environments **2**(3), 311–325 (2010)

[15] Singla, G., Cook, D.J., Schmitter-Edgecombe, M.: Recognizing independent and joint activities among multiple residents in smart environments. Journal of ambient intelligence and humanized computing **1**(1), 57–63 (2010)

[16] Fleury, A., Vacher, M., Noury, N.: Svm-based multimodal classification of activities of daily living in health smart homes: sensors, algorithms, and first experimental results. IEEE transactions on information technology in biomedicine **14**(2), 274–283 (2009)

[17] Cook, D.J., Crandall, A.S., Thomas, B.L., Krishnan, N.C.: Casas: A smart home in a box. Computer **46**(7), 62–69 (2012)

[18] Alemdar, H., Ertan, H., Incel, O.D., Ersoy, C.: Aras human activity datasets in multiple homes with multiple residents. In: 2013 7th International Conference on Pervasive Computing Technologies for Healthcare and Workshops, pp. 232–235 (2013). IEEE

[19] Lago, P., Lang, F., Roncancio, C., Jiménez-Guarín, C., Mateescu, R., Bonnefond, N.: The contextact@ a4h real-life dataset of daily-living activities: Activity recognition using model checking. In: International and Interdisciplinary Conference on Modeling and Using Context, pp. 175–188 (2017). Springer

[20] Alirezaie, M., Renoux, J., Köckemann, U., Kristoffersson, A., Karlsson, L., Blomqvist, E., Tsiftes, N., Voigt, T., Loutfi, A.: An ontology-based context-aware system for smart homes: E-care@ home. Sensors **17**(7), 1586 (2017)

[21] McConville, R., Byrne, D., Craddock, I., Piechocki, R., Pope, J., Santos-Rodriguez, R.: A dataset for room level indoor localization using a smart home in a box. Data in brief **22**, 1044–1051 (2019)

[22] Köckemann, U., Alirezaie, M., Renoux, J., Tsiftes, N., Ahmed, M.U., Morberg, D., Lindén, M., Loutfi, A.: Open-source data collection and data sets for activity recognition in smart homes. Sensors **20**(3), 879 (2020)

[23] Arrotta, L., Bettini, C., Civitarese, G.: The marble dataset: Multi-inhabitant activities of daily living combining wearable and environmental sensors data. In: International Conference on Mobile and Ubiquitous Systems: Computing, Networking, and Services, pp. 451–468 (2021). Springer

[24] Ramos, R.G., Domingo, J.D., Zalama, E., Gómez-García-Bermejo, J., López, J.: Sdhar-home: A sensor dataset for human activity recognition at home. Sensors **22**(21), 8109 (2022)

[25] Tonkin, E.L., Holmes, M., Song, H., Twomey, N., Diethe, T., Kull, M., Perello Nieto, M., Camplani, M., Hannuna, S., Fafoutis, X., *et al.*: A multi-sensor dataset with annotated activities of daily living recorded in a residential setting. Scientific Data **10**(1), 162 (2023)